

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

```
df = pd.read_csv("/content/Titanic-Dataset.csv")
print(df.head(), "\n")
print(df.shape, "\n")
print(df.info())
```

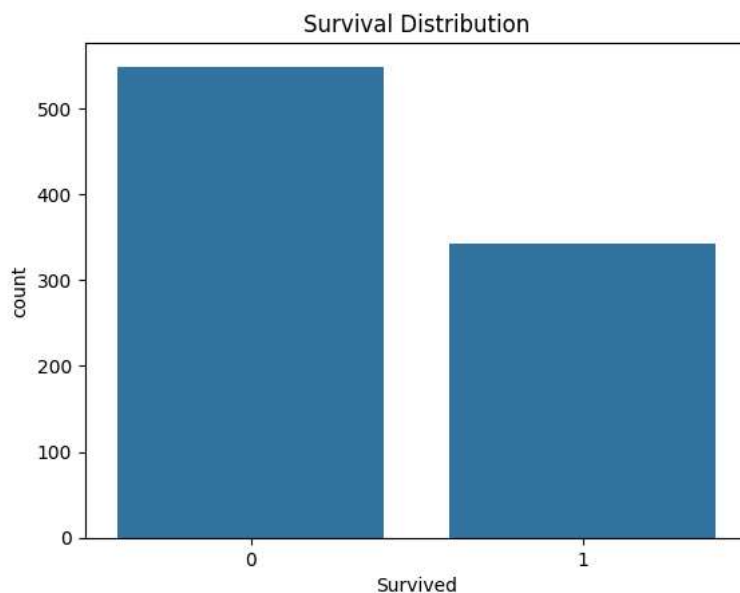
```
   PassengerId  Survived  Pclass  ...    Fare Cabin Embarked
0             1         0       3  ...   7.2500   NaN        S
1             2         1       1  ...  71.2833   C85        C
2             3         1       3  ...   7.9250   NaN        S
3             4         1       1  ...  53.1000  C123        S
4             5         0       3  ...   8.0500   NaN        S
```

```
[5 rows x 12 columns]
```

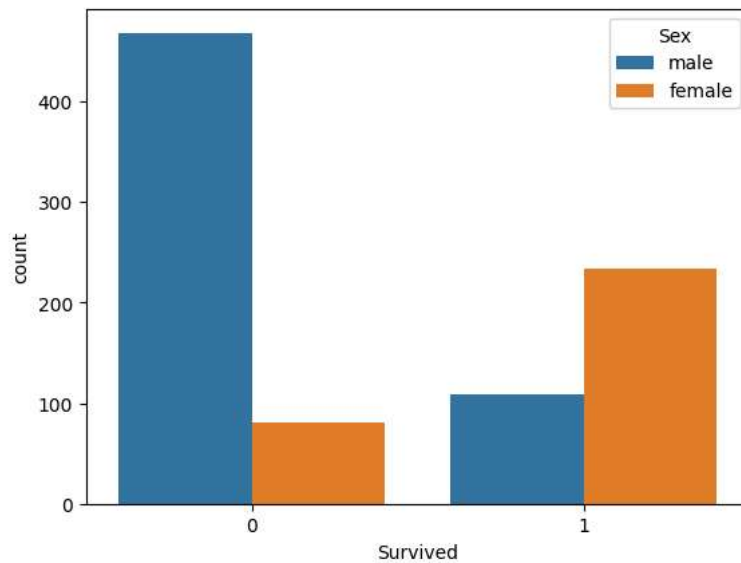
```
(891, 12)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

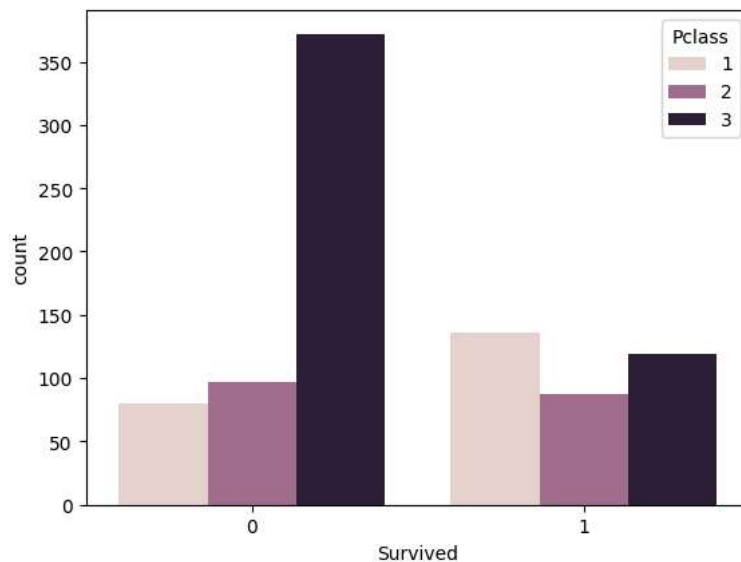
```
sns.countplot(x = 'Survived', data = df)
plt.title("Survival Distribution")
plt.show()
```



```
sns.countplot(x='Survived', hue='Sex', data = df)
plt.show()
```



```
sns.countplot(x='Survived', hue='Pclass', data = df)
plt.show()
```



```
print(df.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

```
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df.drop_duplicates()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.0	1	0	7.2500	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.9250	S
3	1	1	female	35.0	1	0	53.1000	S
4	0	3	male	35.0	0	0	8.0500	S
...	...	...	...	...	...	...	...	...
885	0	3	female	39.0	0	5	29.1250	Q
887	1	1	female	19.0	0	0	30.0000	S
888	0	3	female	28.0	1	2	23.4500	S
889	1	1	male	26.0	0	0	30.0000	C
890	0	3	male	32.0	0	0	7.7500	Q

775 rows × 8 columns

```
df.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'],
axis = 1,
inplace = True)
```

```
print(df.select_dtypes(include='object').columns)

Index(['Sex', 'Embarked'], dtype='object')
```

```
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
```

```
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)
```

```
x = df.drop('Survived',axis=1)
y = df['Survived']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    x,y,
    test_size = 0.2,
    random_state = 42
)
```

```
model = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)

model.fit(X_train, y_train)
```

▼ RandomForestClassifier ⓘ ?

RandomForestClassifier(n_estimators=200, random_state=42)

```
y_pred = model.predict(X_test)
print(y_pred)
print(y_test)
```

```
[0 0 0 1 0 1 1 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 0 1 1 1 0 0 1
 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 1 0 1 0 1 1 0 0 1 1 0 0 0 1 1 1 1 1 1
 0 0 1 1 1 1 0 1 1 0 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 1 1 0 0 1 1
 0 1 1 0 0 1 0 1 0 0 1 1 1 0 0 1 1 0 0 1 0 0 1 1 0 1 0 0 0 0 1 1 0 0 1 0 1
 1 0 0 0 0 1 0 0 0 1 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 1 0 0 0 1 1]
709    1
439    0
840    0
720    1
39     1
..
433    0
773    0
25     1
```

```
84      1
10      1
Name: Survived, Length: 179, dtype: int64
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Accuracy: 0.8100558659217877

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